

Start here X rate_monotonic.c X

```
1  #include <stdio.h>
2  #define MAX 10
3  typedef struct {
4      int id;
5      int execution_time;
6      int period;
7      int remaining_time;
8      int next_deadline;
9  } Task;
10 void sort_by_priority(Task t[], int n) {
11     Task temp;
12     for(int i = 0; i < n - 1; i++) {
13         for(int j = i + 1; j < n; j++) {
14             if(t[i].period > t[j].period) {
15                 temp = t[i];
16                 t[i] = t[j];
17                 t[j] = temp;
18             }
19         }
20     }
21 }
22 void main() {
23     Task t[MAX];
24     int n, hyperperiod = 1, time = 0;
25     printf("Enter number of tasks: ");
26     scanf("%d", &n);
27     for(int i = 0; i < n; i++) {
28         printf("Enter execution time and period for Task %d: ", i + 1);
29         scanf("%d %d", &t[i].execution_time, &t[i].period);
30         t[i].id = i + 1;
31         t[i].remaining_time = 0;
32         t[i].next_deadline = 0;
33     }
34     for(int i = 0; i < n; i++) {
35         hyperperiod *= t[i].period;
36     }
37     sort_by_priority(t, n);
38     printf("\nScheduling Order:\n");
```

Start here x rate monotonic.c x

```
24     int n, hyperperiod = 1, time = 0;
25     printf("Enter number of tasks: ");
26     scanf("%d", &n);
27     for(int i = 0; i < n; i++) {
28         printf("Enter execution time and period for Task %d: ", i + 1);
29         scanf("%d %d", &t[i].execution_time, &t[i].period);
30         t[i].id = i + 1;
31         t[i].remaining_time = 0;
32         t[i].next_deadline = 0;
33     }
34     for(int i = 0; i < n; i++) {
35         hyperperiod *= t[i].period;
36     }
37     sort_by_priority(t, n);
38     printf("\nScheduling Order:\n");
39     for(time = 0; time < hyperperiod; time++) {
40         for(int i = 0; i < n; i++) {
41             if(time % t[i].period == 0) {
42                 t[i].remaining_time = t[i].execution_time;
43                 t[i].next_deadline = time + t[i].period;
44             }
45         }
46         int selected = -1;
47         for(int i = 0; i < n; i++) {
48             if(t[i].remaining_time > 0) {
49                 selected = i;
50                 break;
51             }
52         }
53         if(selected != -1) {
54             printf("Time %d -> Task %d\n", time, t[selected].id);
55             t[selected].remaining_time--;
56         } else {
57             printf("Time %d -> Idle\n", time);
58         }
59     }
60 }
61
```

```
"C:\Users\BMSCECSE-L3-04\D  X + v
Enter execution time and period for Task 1: 10
5
Enter execution time and period for Task 2: 15
4

Scheduling Order:
Time 0 -> Task 2
Time 1 -> Task 2
Time 2 -> Task 2
Time 3 -> Task 2
Time 4 -> Task 2
Time 5 -> Task 2
Time 6 -> Task 2
Time 7 -> Task 2
Time 8 -> Task 2
Time 9 -> Task 2
Time 10 -> Task 2
Time 11 -> Task 2
Time 12 -> Task 2
Time 13 -> Task 2
Time 14 -> Task 2
Time 15 -> Task 2
Time 16 -> Task 2
Time 17 -> Task 2
Time 18 -> Task 2
Time 19 -> Task 2

Process returned 20 (0x14)    execution time : 12.832 s
Press any key to continue.
```

"C:\Users\BMSCECSE-L3-04\D X

+ v

- □ X

Enter execution time and period for Task 1: 1

4

Enter execution time and period for Task 2: 2

5

Scheduling Order:

Time 0 -> Task 1

Time 1 -> Task 2

Time 2 -> Task 2

Time 3 -> Idle

Time 4 -> Task 1

Time 5 -> Task 2

Time 6 -> Task 2

Time 7 -> Idle

Time 8 -> Task 1

Time 9 -> Idle

Time 10 -> Task 2

Time 11 -> Task 2

Time 12 -> Task 1

Time 13 -> Idle

Time 14 -> Idle

Time 15 -> Task 2

Time 16 -> Task 1

Time 17 -> Task 2

Time 18 -> Idle

Time 19 -> Idle

Process returned 20 (0x14) execution time : 5.283 s

Press any key to continue.

Start here X rate monotonic.c X earliest deadline first.c X

```
1  #include <stdio.h>
2  #define MAX 10
3  typedef struct {
4      int id;
5      int execution_time;
6      int period;
7      int remaining_time;
8      int deadline;
9  } Task;
10
11 int main() {
12     Task t[MAX];
13     int n, time, hyperperiod = 1;
14
15     printf("Enter number of tasks: ");
16     scanf("%d", &n);
17
18     for(int i = 0; i < n; i++) {
19         printf("Enter execution time and period for Task %d: ", i + 1);
20         scanf("%d %d", &t[i].execution_time, &t[i].period);
21
22         t[i].id = i + 1;
23         t[i].remaining_time = 0;
24         t[i].deadline = 0;
25     }
26
27     for(int i = 0; i < n; i++) {
28         hyperperiod *= t[i].period;
29     }
30
31     printf("\nEarliest Deadline First Scheduling Order:\n");
32
33     for(time = 0; time < hyperperiod; time++) {
34
35         for(int i = 0; i < n; i++) {
36             if(time % t[i].period == 0) {
37                 t[i].remaining_time = t[i].execution_time;
38                 t[i].deadline = time + t[i].period;
```

Start here x rate monotonic.c x earliest deadline first.c x

```
25     }
26
27     for(int i = 0; i < n; i++) {
28         hyperperiod *= t[i].period;
29     }
30
31     printf("\nEarliest Deadline First Scheduling Order:\n");
32
33     for(time = 0; time < hyperperiod; time++) {
34
35         for(int i = 0; i < n; i++) {
36             if(time % t[i].period == 0) {
37                 t[i].remaining_time = t[i].execution_time;
38                 t[i].deadline = time + t[i].period;
39             }
40         }
41
42         int selected = -1;
43         int earliest_deadline = 1e9;
44
45         for(int i = 0; i < n; i++) {
46             if(t[i].remaining_time > 0 && t[i].deadline < earliest_deadline) {
47                 earliest_deadline = t[i].deadline;
48                 selected = i;
49             }
50         }
51
52         if(selected != -1) {
53             printf("Time %d -> Task %d\n", time, t[selected].id);
54             t[selected].remaining_time--;
55         } else {
56             printf("Time %d -> Idle\n", time);
57         }
58     }
59
60     return 0;
61 }
62
```

```
"C:\Users\BMSCECSE-L3-04\D  X + v
Enter execution time and period for Task 1: 1
4
Enter execution time and period for Task 2: 2
5

Earliest Deadline First Scheduling Order:
Time 0 -> Task 1
Time 1 -> Task 2
Time 2 -> Task 2
Time 3 -> Idle
Time 4 -> Task 1
Time 5 -> Task 2
Time 6 -> Task 2
Time 7 -> Idle
Time 8 -> Task 1
Time 9 -> Idle
Time 10 -> Task 2
Time 11 -> Task 2
Time 12 -> Task 1
Time 13 -> Idle
Time 14 -> Idle
Time 15 -> Task 2
Time 16 -> Task 1
Time 17 -> Task 2
Time 18 -> Idle
Time 19 -> Idle

Process returned 0 (0x0)   execution time : 7.319 s
Press any key to continue.
```

Start here x rate_monotonic.c x *earliest deadline first.c x proportional.c x

```
1  #include<stdio.h>
2  #define MAX 10
3  typedef struct {
4      int id;
5      int weight;
6      int execution_time;
7      int remaining_time;
8      float cpu_received;
9  } Task;
10 void main() {
11     Task t[MAX];
12     int n, total_time = 0;
13     printf("Enter number of tasks: ");
14     scanf("%d", &n);
15     for(int i = 0; i < n; i++) {
16         printf("Enter execution time and weight for Task %d: ", i + 1);
17         scanf("%d %d", &t[i].execution_time, &t[i].weight);
18         t[i].id = i + 1;
19         t[i].remaining_time = t[i].execution_time;
20         t[i].cpu_received = 0.0;
21         total_time += t[i].execution_time;
22     }
23     printf("\nProportional Scheduling Order:\n");
24     for(int time = 0; time < total_time; time++) {
25         int selected = -1;
26         float min_ratio = 1e9;
27         for(int i = 0; i < n; i++) {
28             if(t[i].remaining_time > 0) {
29                 float ratio = t[i].cpu_received / t[i].weight;
30                 if(ratio < min_ratio) {
31                     min_ratio = ratio;
32                     selected = i;
33                 }
34             }
35         }
36         if(selected != -1) {
37             printf("Time %d -> Task %d\n", time, t[selected].id);
38
39             t[selected].remaining_time--;
40             t[selected].cpu_received += 1.0;
41         }
42     }
43 }
44
```



```
"C:\Users\BMSCECSE-L3-04\D  X + v
Enter number of tasks: 2
Enter execution time and weight for Task 1: 4
1
Enter execution time and weight for Task 2: 4
2
Proportional Scheduling Order:
Time 0 -> Task 1
Time 1 -> Task 2
Time 2 -> Task 2
Time 3 -> Task 1
Time 4 -> Task 2
Time 5 -> Task 2
Time 6 -> Task 1
Time 7 -> Task 1
Process returned 8 (0x8)  execution time : 8.791 s
Press any key to continue.
|
```